

Mathematics

A Cultural Heritage

Lecture 4

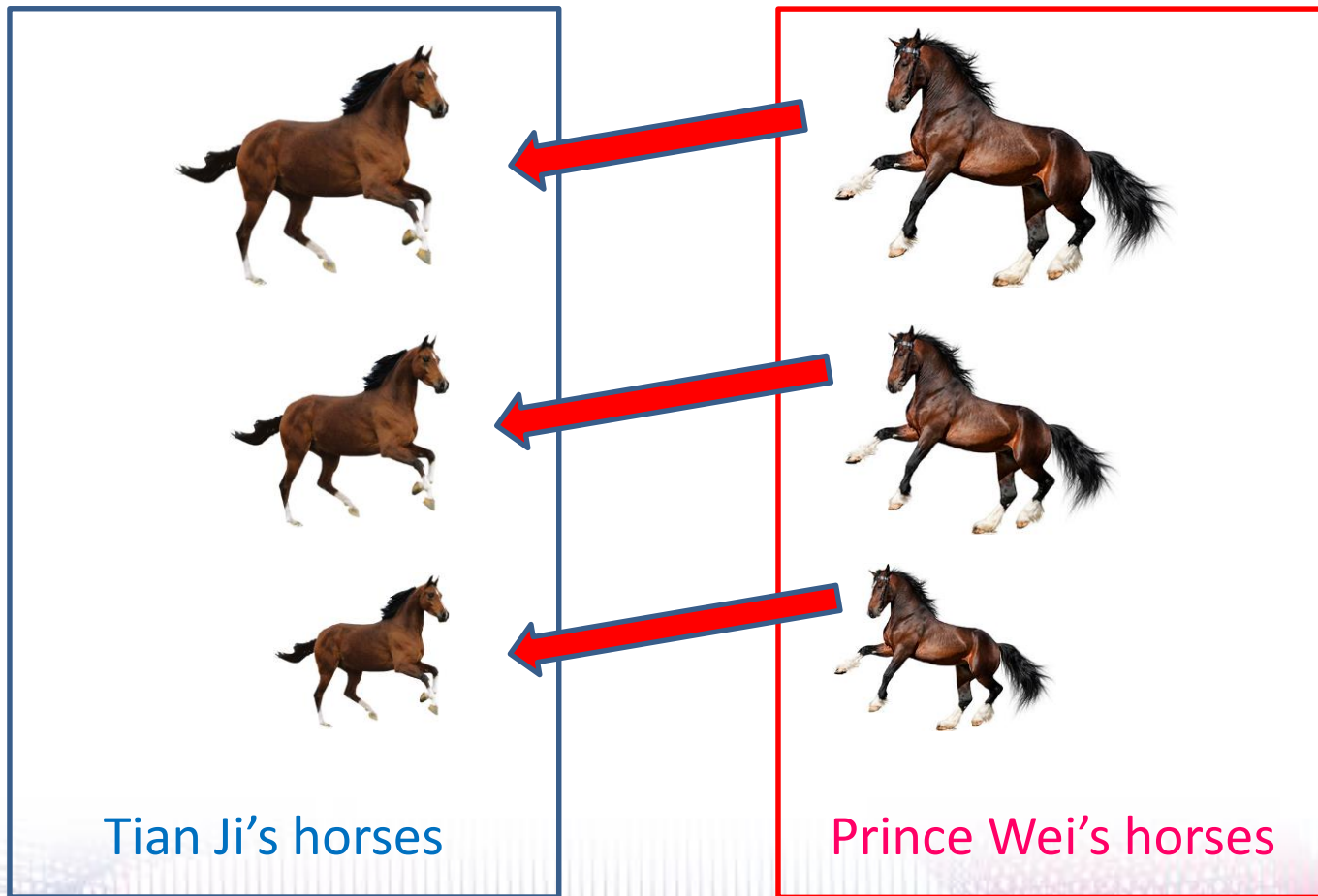
New Kinds of Math – Part 4: Game Theory

Game Theory

- Game Theory = Decision making
- Horse Race of **Tian Ji** (田忌賽馬) in ancient China (during the Warring States period):
 - **Prince Wei** (齊威王, ~378 – 320 BC) of the Qi State and his General **Tian Ji** (田忌) have horse races.
 - Top, Middle, and Low class horses

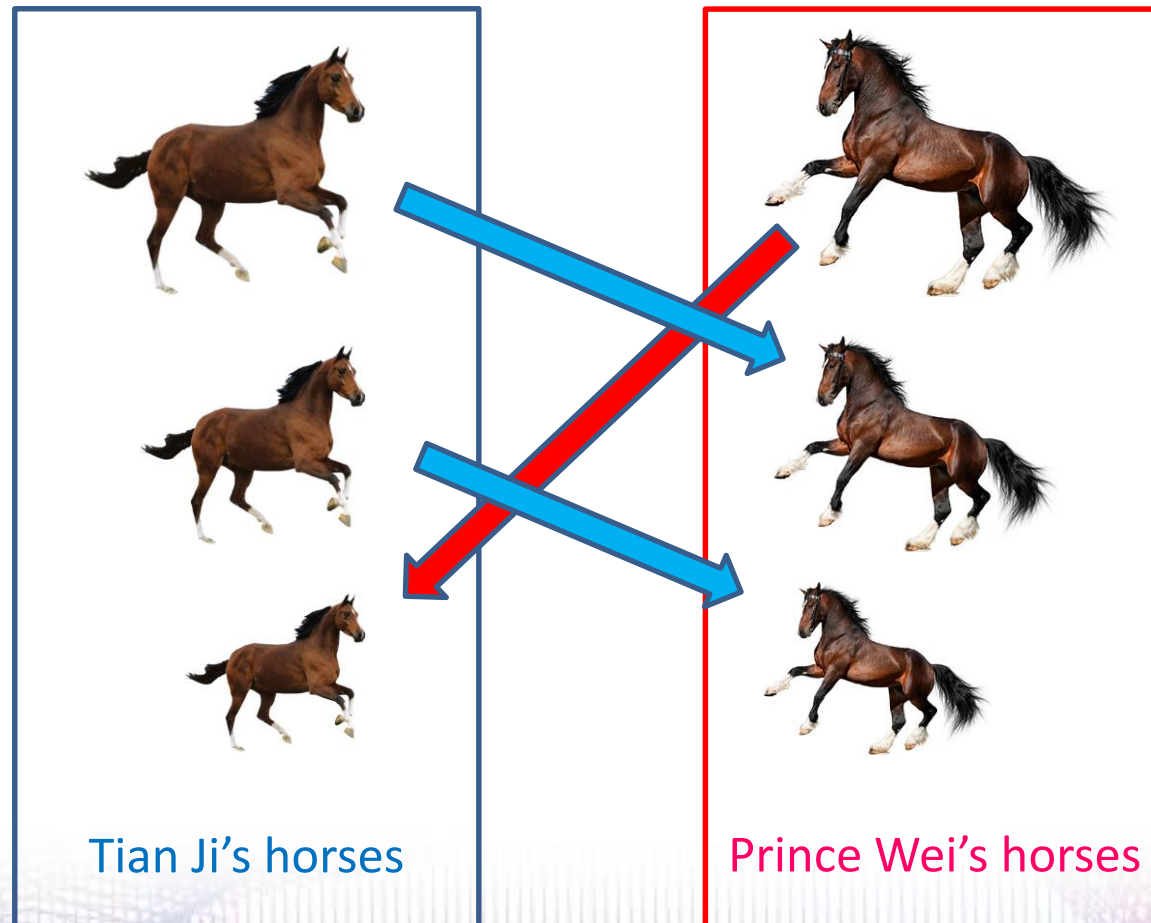
Tian Ji's Horse Race

1. Each of Tian Ji's horses is worse:



A Clever Strategy

1. Sun Bin (孫臏) offers strategy:



Sun Bin

Mathematical Strategies in Modern Times

- Von Neumann (1928): Following games:
 - two-person,
 - zero-sum,
 - perfect-information

Can minimize one's maximum (**minimax**) lost.

John von Neumann

1. John von Neuman (1903-1957)

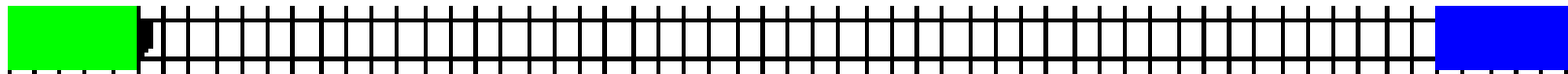
- Born in Hungary, emigrated to US: 1930.
- Mathematician, Physicist, computer scientist, ...
- Pioneer in many fields: math, physics, computer science, economics, artificial intelligence, statistics, ...
- Can do complicated computations in head, and has photographic memory.



John von Neumann

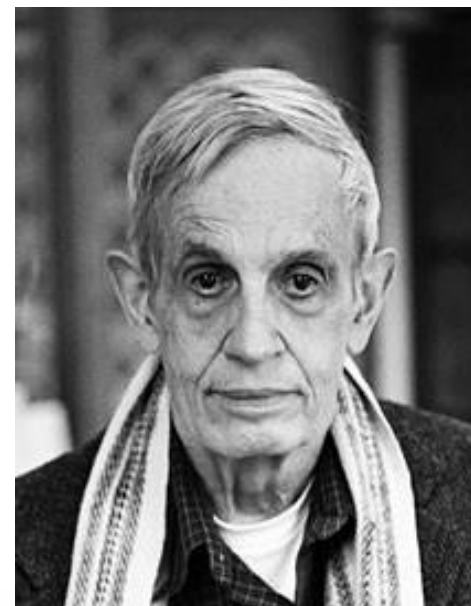
The Flying Bee Problem

1. Two **trains**, **200 km** apart, are moving towards each other, each with speed **50 km/hr**.
2. A **bee** starts at the front of train *A* and flies towards train *B* at **75 km/hr**; as soon as it reaches the front of the other train, it will turn around to fly in the opposite direction. This is repeated until the two trains collide.
3. **Question**: how far would the bee have travelled?



Further Generalizations

- The 1928 result of von Neumann later improved and extended many times, e.g., to imperfect information games, or games with more than two players.
- A significant extension of the result is the concept of **Nash equilibrium**, by **John Nash** in 1950.
- Nash got the Nobel prize (in economics) for this in 1994.



John Nash
(1928 –2015)

The Prisoner's Dilemma

- Two criminals were arrested:
 - Each is locked in a cellar, without means of communication to the other.
 - If both keep silent, both will get **1** year of jail term.
 - If both betray, both will get **2** years of jail term.
 - If *A* keeps silent but *B* betrays, *A* will get **3** years jail term, and *B* can be **free** (and vice versa).
- Assume both criminals are rational. What would each of them do to benefit him/herself?

The Prisoner's Dilemma

1. Consider the following **pay-off matrix**:

	<i>B</i> keeps silent	<i>B</i> betrays
<i>A</i> keeps silent	Each serves 1 year	<i>A</i> serves 3 years <i>B</i> free
<i>A</i> betrays	<i>A</i> free <i>B</i> serves 3 years	Each serves 2 years